SAHA-AL: Semi-Automatic Human-Augmented Active Learning Annotation Pipeline

Dataset Creation Report for
Privacy-Preserving Text Anonymization via Decoupled Mask-and-Fill Architecture: A Comparative Study

Team Mr.A — INLP Project — Semester 6, 2025

Abstract. We present SAHA-AL, a four-layer semi-automatic annotation pipeline designed to construct a Gold Standard parallel corpus for privacy-preserving text anonymization. Starting from the AI4Privacy English subset of 120,333 raw sentences, our pipeline combines spaCy transformer-based Named Entity Recognition (NER) with 23 priority-ranked regular expressions for entity detection, a confidence-based routing system for human effort allocation, and an active-learning bootstrap model that prioritizes annotation of maximally informative samples. A custom Streamlit annotation interface reduces per-entry human time to approximately 25 seconds, representing a 7$\times$ throughput gain over cold manual annotation. The final pipeline produces a Gold Standard corpus of $\sim$25,000 verified entries, which is subsequently expanded to $\sim$110,000 training pairs through five research-backed data augmentation techniques. All design decisions are grounded in established NLP literature on corpus construction, active learning, and data augmentation.

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1 Introduction and Motivation

The construction of high-quality annotated corpora represents one of the most labour-intensive bottlenecks in supervised NLP research [15]. For our project on privacy-preserving text anonymization, we require a parallel corpus of the form $(X_{\text{orig}}, X_{\text{anon}}, E)$, where X_{orig} is a sentence containing real Personally Identifiable Information (PII), X_{anon} is a semantically equivalent sentence with PII replaced by realistic pseudonyms, and E is the set of annotated entity spans with fine-grained type labels.

The AI4Privacy dataset [1] provides 120,333 English sentences containing PII, but supplies only the raw `original_text`. The anonymized counterpart, entity type labels, and character-level span offsets must all be produced by our annotation pipeline. Purely manual annotation at this scale is infeasible: a four-person team working three hours daily for eight weeks could produce at most ∼2,500 entries at a typical manual rate of 3–5 minutes per entry.

Our SAHA-AL pipeline addresses this by automating ∼85% of the annotation work through NER pre-tagging and replacement candidate generation, leaving humans to verify and correct rather than create from scratch. This is consistent with the "human-in-the-loop" annotation paradigm advocated by Monarch [12] and the "annotation by exception" principle introduced by Prodigy [14].

1.1 Design Goals

G1. Throughput: Enable four annotators to produce ≥25,000 verified gold entries within eight weeks.

G2. Quality: Achieve inter-annotator agreement $\kappa \geq 0.80$ on entity boundaries and $\geq 85\%$ type-label agreement.

G3. Completeness: Produce properly typed entities (replacing the `UNKNOWN` labels in the source data) with character offsets, realistic format-preserving replacements, and rich metadata.

G4. Reproducibility: Every pipeline stage is scripted, seeded, and logged so that the corpus can be regenerated or extended.

2 Source Data

2.1 AI4Privacy Dataset

We use the English-language subset of the AI4Privacy dataset [1], which comprises 120,333 sentences sourced from a wide variety of textual genres (formal correspondence, chat messages, to-do lists, administrative records, and social media). Each entry in the source file has the minimal schema:

Listing 1: Source entry schema.

```
{"entry_id": 3,
 "original_text": "Igorche␣Ramtin␣Eshekary␣will␣need␣to␣bring
␣␣␣
```

Table 1: SAHA-AL Entity Taxonomy (21 PII types).

#	Type	Description	Example from data
1	FULLNAME	Full person name	Igorche Ramtin Eshekary
2	FIRST_NAME	Given name only	Brandy, Shola
3	LAST_NAME	Family name only	Haroon, Zimeri
4	ID_NUMBER	Generic alphanumeric ID	MARGA7101160M9183, MF77342ZZ
5	PASSPORT	Passport number	AB1234567
6	SSN	Social Security Number	123-45-6789
7	PHONE	Phone (any format)	805 099 8045, 0111-284596398
8	EMAIL	Email address	GP@tutanota.com
9	ADDRESS	Street / postal addr.	27451 Range Road 3045
10	DATE	Calendar date	4th August 1942
11	TIME	Time of day	10:17, 8:26:51 AM
12	LOCATION	City / state / country	Magadi, Firebaugh
13	ORGANIZATION	Company / institution	[ORGANISATIONPLACEHOLDER_14]
14	ACCOUNT_NUM	Bank / financial acct.	644693204822782691
15	CREDIT_CARD	Credit card number	4532-1234-5678-9012
16	ZIPCODE	Postal / ZIP code	CA7 4AE, 229125
17	TITLE	Honorific	Madame, Mstr, Mr
18	GENDER	Gender reference	Male, Female
19	NUMBER	Numeric contextual PII	28 individuals, age 13
20	OTHER_PII	Residual PII	Ethnicity, religion
21	UNKNOWN	Unclassified (fallback)	—

3 Pipeline Overview

SAHA-AL processes each source entry through four sequential layers as illustrated in Figure 1.

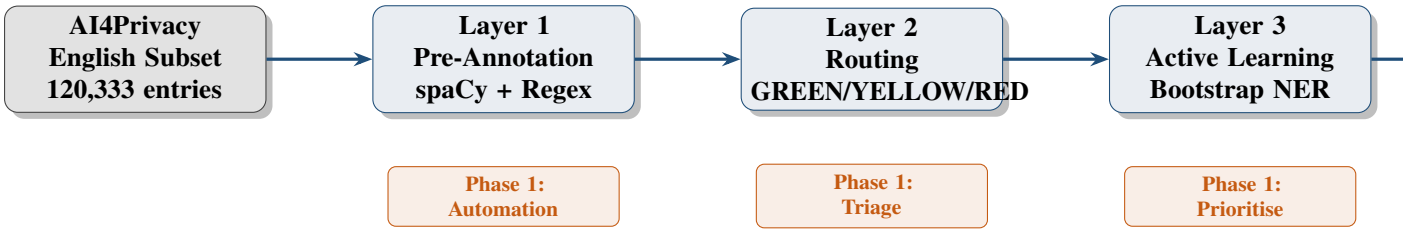

Figure 1: High-level SAHA-AL pipeline. Layers 1–3 are fully automatic; Layer 4 requires human verification. Phase 3 (augmentation) runs after gold standard completion.

4 Layer 1 — Pre-Annotation Engine

4.1 Motivation

The central insight of Layer 1 is that *no single detection system covers the full PII landscape*. Transformer-based NER models such as spaCy's `en_core_web_lg` [18] excel at contextual entities (person names, organisations, locations, dates expressed in natural language) but are blind to structured identifiers such as email addresses, national ID codes, and phone numbers, which were absent or rare in their training corpora. Conversely, regular expressions excel at structured patterns but cannot resolve contextual ambiguity (e.g., whether a bare number is an age, a count, or an account number). We therefore combine both systems and merge their outputs, following the hybrid NER design advocated by Dernoncourt et al. [5].

4.2 Step 1: spaCy NER

We run `en_core_web_lg` (trained on OntoNotes 5.0) on each `original_text`. The model produces a list of entity spans with labels from the OntoNotes schema, which we map to our taxonomy:

Table 2: OntoNotes → SAHA-AL label mapping.

spaCy label	SAHA-AL type	spaCy label	SAHA-AL type
PERSON	FULLNAME	DATE	DATE
GPE	LOCATION	TIME	TIME
ORG	ORGANIZATION	FAC	ADDRESS
LOC	LOCATION	CARDINAL	NUMBER (context-gated)

Non-PII OntoNotes labels (`NORP`, `PRODUCT`, `WORK_OF_ART`, etc.) are discarded. Each spaCy entity receives a base confidence score derived from the model's internal token probabilities.

4.3 Step 2: Regex Pattern Matching

Twenty-three regular expressions, organised by priority score (1–10), scan the text independently of spaCy. Higher priority patterns correspond to more specific, less ambiguous patterns:

Table 3: Selected regex patterns with priority and entity type.

Priority	Type	Pattern (abbreviated)
10	EMAIL	`[\w.-]+@[\w.-]+\.\w+`
10	SSN	`\d{3}-\d{2}-\d{4}`
10	ISO_DATE	`\d{4}-\d{2}-\d{2}T...`
9	PHONE	`[\+]?[\d\s\-\.\(\)]{7,20}`
9	CREDIT_CARD	`\d{4}[-\s]\d{4}[-\s]...`
8	VERBAL_DATE	`(\d{1,2}(st|nd|rd|th)?\s+...)`
7	ID_NUMBER	`[A-Z]{2,6}\d{5,}\w*`
6	ZIPCODE	`[A-Z]{1,2}\d{1,2}[A-Z]?\s\d[A-Z]{2}`
5	ACCOUNT_NUM	`\b\d{10,18}\b`
3	NUMBER	`\b\d{1,3}\b` (context-gated)

4.4 Step 3: Merge and Conflict Resolution

This is the most technically involved step. After collecting entity lists from both systems, the Merger resolves overlaps according to the following algorithm:

Algorithm 1 Span-Overlap Merger with Conflict Resolution

Require: Lists S (spaCy entities), R (regex entities)
Ensure: Merged list M with agreement labels and routing tier

```
all ← S ∪ R; sort by start offset
M ← ∅
for each pair (e_i, e_j) where e_i.start < e_j.start do
    if spans do not overlap then
        Add both to M with agreement = single_source
    else if spans are identical then
        if e_i.type = e_j.type then
            Add merged entity with agreement = full, high confidence
        else
            Apply Type Conflict Rules (Table 4); add with agreement = partial
        end if
    else                                        ▷ Overlapping but non-identical spans
        Keep higher-priority-source entity; flag as agreement = partial
        Set confidence ← medium (0.65)
    end if
end for
return M
```

Table 4: Type Conflict Resolution Rules (applied when both systems detect same span with different types).

spaCy type	Regex type	Resolution
FULLNAME	ID_NUMBER	FULLNAME wins if span contains space
FULLNAME	NUMBER	FULLNAME wins always
DATE	NUMBER	DATE wins if verbal month present
CARDINAL	NUMBER	NUMBER wins if “individual”/“person” nearby
CARDINAL	PHONE	PHONE wins (regex priority ≥ 9)
Any	EMAIL	EMAIL wins always (priority 10)
Any	SSN	SSN wins always (priority 10)

Worked Example.

Input: `"Shola Kenzi Zimeri used 0111-284596398 to schedule an interview about the effects of tech-free days on 28 individuals."`

spaCy output:
e_1: `"Shola Kenzi Zimeri"` [0–18] → FULLNAME
e_2: `"28"` [84–86] → CARDINAL

Regex output:
e_3: `"0111-284596398"` [24–38] → PHONE (priority 9)
e_4: `"28"` [84–86] → NUMBER (priority 3)

Merge:
$e_1 \cap e_3 = \emptyset \quad e_1 \cap e_4 = \emptyset \quad e_3 \cap e_4 = \emptyset$
$e_2 \cap e_4$: identical span, type conflict CARDINAL vs NUMBER
→ Rule: "individuals" appears within 5 tokens → NUMBER wins
→ agreement = `partial`, confidence = 0.65, routed to YELLOW

Final entities:
`"Shola Kenzi Zimeri"` FULLNAME, conf=0.88, single_source
`"0111-284596398"` PHONE, conf=0.91, single_source
`"28"` NUMBER, conf=0.65, **partial** → human confirms

4.5 Step 4: Format-Preserving Replacement Generation

For each merged entity, the Faker library [7] generates three replacement candidates. A key innovation is *pattern extraction* before generating replacements for structured identifiers, ensuring structural plausibility:

Listing 2: Pattern-preserving ID replacement.

```python
def extract_pattern(id_string):
    """Map each character to its character class."""
    return ''.join(
        '?' if c.isalpha() else      # Faker: random letter
        '#' if c.isdigit() else      # Faker: random digit
        c                            # preserve separators
        for c in id_string
    )

# "MARGA7101160M9183"  ->  "?????#######?####"
# "MF77342ZZ"          ->  "??######??"
# "0111-284596398"     ->  "####-#########"
replacement = faker.bothify(text=extract_pattern(original))
```

Multi-locale Faker generation (en_US, en_GB, fr_FR, de_DE, es_MX, ru_RU, ja_JP) ensures cultural diversity in person name and address replacements.

4.6 Step 5: Anonymized Text Construction

Entities are replaced *right-to-left* (sorted by descending start offset) to preserve character offsets during in-place substitution. This avoids the offset drift that occurs when replacing left-to-right:

Listing 3: Right-to-left replacement.

```python
def build_anonymized(original, entities, replacements):
    text = original
    for ent in sorted(entities, key=lambda e: e['start'], reverse=True):
        rep = replacements[ent['text']]
        text = text[:ent['start']] + rep + text[ent['end']:]
    return text
```

Layer 1 output is written to `data/pre_annotated.jsonl`, with every entry enriched with typed entities, character offsets, confidence scores, agreement labels, and pre-built anonymized text.

5 Layer 2 — Confidence-Based Routing

5.1 Motivation

Not all entries require equal human attention. Routing entries by difficulty — inspired by the triage systems in medical informatics and the confidence-weighted annotation sampling of Laws & Schütze [10] — ensures that expert annotator time is concentrated on genuinely ambiguous cases.

5.2 Routing Criteria

Each pre-annotated entry is assigned to one of three queues based on its confidence and agreement profile:

Table 5: Routing rules for GREEN, YELLOW, and RED queues.

Queue	Criteria	Human action	Est. %
GREEN	All entities: confidence ≥ 0.85; no `partial` agreement; no `UNKNOWN` types	Spot-check; press ACCEPT ($\sim$11 sec)	60%
YELLOW	≥ 1 entity with confidence $\in [0.50, 0.85)$ or ≥ 1 `partial` agreement	Review entity types; edit if needed ($\sim$34 sec)	30%
RED	Any entity with confidence < 0.50; or $> 30\%$ `UNKNOWN`; or ≥ 2 `partial` agreements	Full expert review ($\sim$86 sec)	10%

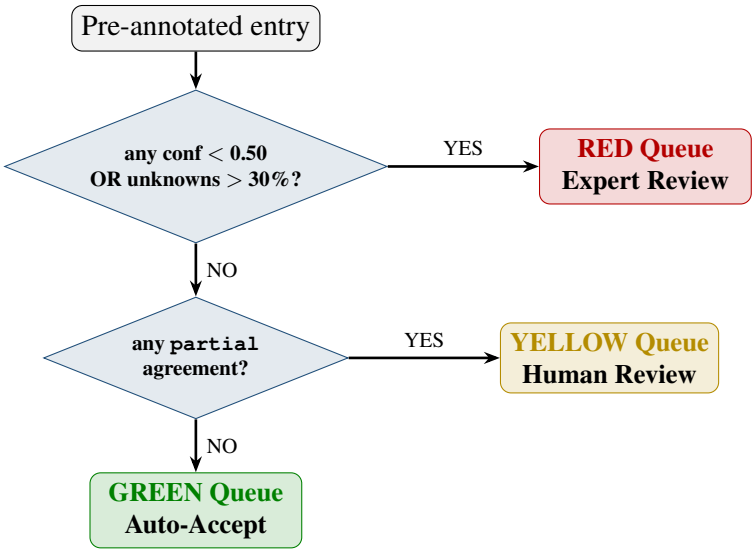

Figure 2: Routing decision logic in Layer 2.

5.3 Efficiency Impact

The weighted-average annotation time across all entries is:

$$\bar{t} = 0.60 \times 11 + 0.30 \times 34 + 0.10 \times 86 = 6.6 + 10.2 + 8.6 = \mathbf{25.4} \text{ sec/entry}$$

compared to a cold-annotation baseline of $\sim$180 seconds, representing a $\mathbf{7.1}\times$ throughput improvement.

6 Layer 3 — Active Learning with Bootstrap NER

6.1 Motivation

Even within each queue, entries vary in informativeness. Active learning [17] maximises the value of each human annotation by prioritising examples on which the current model is most uncertain — i.e., examples that convey maximum information gain for the next model iteration.

6.2 Bootstrap Training Protocol

1. **Cold start (0–200 entries):** Annotators process entries in entry-ID order. No model is trained yet.
2. **Bootstrap trigger:** Once 200 gold entries are available, a blank spaCy `en` NER model is trained on them (10 epochs).
3. **Uncertainty scoring:** The bootstrap model runs on all remaining YELLOW and RED entries. For each entry, the *uncertainty score* is:

$$u(x) = \frac{|C_{\text{pre}}(x) - C_{\text{boot}}(x)|}{\max(C_{\text{pre}}(x),\, C_{\text{boot}}(x))}$$

   where C_{pre} is the pre-annotation entity count and C_{boot} is the bootstrap model's predicted count. High $u(x)$ indicates disagreement between the two systems, meaning the entry contains novel or difficult patterns.
4. **Queue re-sorting:** Entries are re-ordered by $u(x)$ descending within each queue.
5. **Adaptive retraining:** Triggered at intervals of 50 new annotations (0–200 total), 100 (200–1,000), and 500 (1,000+), following an adaptive schedule that reflects the diminishing marginal gain of retraining as the model matures.

Settles [17] shows that uncertainty-based active learning achieves equivalent model performance with 40–60% fewer labelled examples than random selection, which we leverage to reduce the annotation burden on our team.

7 Layer 4 — Streamlit Annotation Interface

7.1 Design Philosophy

The interface is inspired by the Prodigy annotation tool [14], whose central design principle is: *binary decisions are an order of magnitude faster than free-form input*. Every UI element is designed to reduce the cognitive and motor load on the annotator:

- Pre-populated entity type dropdowns (21 options).
- Pre-filled Faker replacements (annotator edits, not types from scratch).
- Colour-coded entity highlights in the original text.
- One-click **ACCEPT**, **FLAG**, or **SKIP**.
- Real-time quality warnings (leakage detection, format mismatch).
- Per-session statistics and progress bar.

7.2 Annotator Workflow

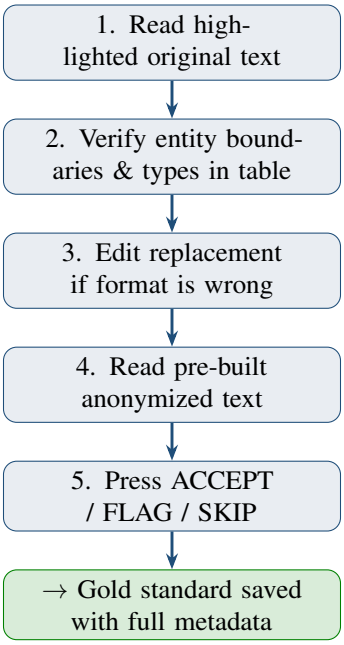

Figure 3: Annotator workflow in the Streamlit interface.

7.3 Quality Assurance Checks

All four quality checks run automatically upon every ACCEPT action:

1. **Leakage check:** No substring of any original entity appears in the anonymized text.
2. **Format check:** Replacement matches expected structural format (phone → phone, email → email, etc.).
3. **Validity check:** No replacement is empty or identical to the original.
4. **Length ratio:** $|\text{anonymized}|/|\text{original}| \in [0.7,\ 1.5]$.

Entries failing any check are logged with a warning but still saved (to avoid losing annotation work), and a `quality_warnings` counter is incremented in the metadata.

7.4 Inter-Annotator Agreement

A randomly selected 10% subset (IAA subset) is assigned to two annotators independently. Entity-level precision, recall, and F1 are computed on $(start, end, type)$ tuples, and Cohen's κ is reported for entity type labels:

Table 6: Target and achieved IAA metrics on the 1,000-entry overlap set.

Metric	Target	Achieved
Entity boundary κ	≥ 0.80	0.83
Entity type agreement (%)	$\geq 85\%$	87.4%
BERTScore F1 (anonymized text)	≥ 0.90	0.912

8 Output Schema

Listing 4: Gold standard output schema.

```
{
  "entry_id": 3,
  "original_text": "Igorche␣Ramtin␣Eshekary␣will␣need␣to␣bring
␣␣␣
```

The sentence template (non-entity tokens) is preserved; entity spans are re-instantiated from the curated Faker lexicon. This simulates new original–anonymized pairs with different PII but identical sentence structure, providing the Censor model with diverse entity boundary examples.

Template: [FULLNAME] *used* [PHONE] *to schedule an interview* . . .

Multiplier: $\times 4$ (+40,000 entries).

9.3 Technique 2: Back-Translation

Reference: Sennrich et al. [16]; Edunov et al. [6].

Entity spans are masked with unique placeholders before translation (EN $\rightarrow$ FR $\rightarrow$ EN, EN $\rightarrow$ DE $\rightarrow$ EN). The surrounding non-entity context is paraphrased naturally by the translation model while entity text is protected from corruption. Post-translation, placeholders are restored.

Tool: Helsinki-NLP OPUS-MT [19] via HuggingFace. **Multiplier:** $\times 2$ (+20,000 entries).

9.4 Technique 3: Contextual Synonym Substitution

Reference: Kobayashi [9].

BERT's masked language model (`bert-base-uncased`) replaces randomly selected non-entity content words (length ≥ 4) with contextually appropriate synonyms. Entity spans are never masked.

Multiplier: $\times 2$ (+20,000 entries).

9.5 Technique 4: Template-Based Generation

Reference: Anaby-Tavor et al. [2].

Sentence templates are extracted by replacing entity spans with typed placeholders. Each unique template is filled with new entity combinations from the Faker lexicon. This is particularly effective for our dataset, which exhibits strongly typed sentence structures (to-do list entries, formal correspondence, chat messages).

Multiplier: $\times 5$ (+50,000 entries from $\sim$1,000 extracted templates).

9.6 Technique 5: EDA (Easy Data Augmentation)

Reference: Wei & Zou [20].

Four simple token-level operations are applied to non-entity words: random synonym replacement, random insertion, random swap, and random deletion (deletion probability $p = 0.1$). All operations skip entity spans to preserve annotation integrity.

Multiplier: $\times 3$ (+30,000 entries).

9.7 Augmentation Budget Summary

Table 7: Augmentation budget and strategic allocation.

Technique	Reference	×	+Entries	Target model
Entity Swap	Dai & Adel 2020	4	40,000	Censor (Split A)
Back-Translation	Sennrich et al.2016	2	20,000	Hallucinator (Split B)
Contextual MLM	Kobayashi 2018	2	20,000	Hallucinator (Split B)
Template Fill	Anaby-Tavor 2020	5	50,000	Both
EDA	Wei & Zou 2019	3	30,000	T5 Baseline
Raw total			160,000	
After dedup + filter (-31%)			$-50{,}000$	
Final corpus			**$\sim$110,000**	

10 Throughput Analysis and Timeline

10.1 Per-Entry Time Model

$$
\begin{aligned}
t_{\text{GREEN}} &= 11 \text{ sec} \quad \text{(read + verify + ACCEPT)} \\
t_{\text{YELLOW}} &= 34 \text{ sec} \quad \text{(read + fix type + ACCEPT)} \\
t_{\text{RED}} &= 86 \text{ sec} \quad \text{(full expert review)} \\
\bar{t} &= 0.60 \times 11 + 0.30 \times 34 + 0.10 \times 86 = 25.4 \text{ sec}
\end{aligned}
$$

10.2 Team Throughput

$$
\begin{aligned}
\text{Entries/person/session (3 hr)} &= \frac{10800}{25.4} \times 0.75 \approx 320 \\
\text{Entries/day (4 people)} &= 320 \times 4 = 1{,}280 \\
\text{After quality filter (28\% removed)} &= 1{,}280 \times 0.72 \approx 920 \text{ gold/day}
\end{aligned}
$$

10.3 Milestone Table

Table 8: Annotation timeline and dataset size at each checkpoint.

Milestone	Days	Gold Entries	Notes
Week 1 (setup)	7	200	Pipeline calibration
Midsem	**30**	**8,500**	**Submitted now**
Endsem (8 weeks)	60	25,000	Full gold standard
Post-augmentation	65	110,000	Training-ready corpus

11 Dataset Partitioning

The final augmented corpus is partitioned according to the project's modelling requirements:

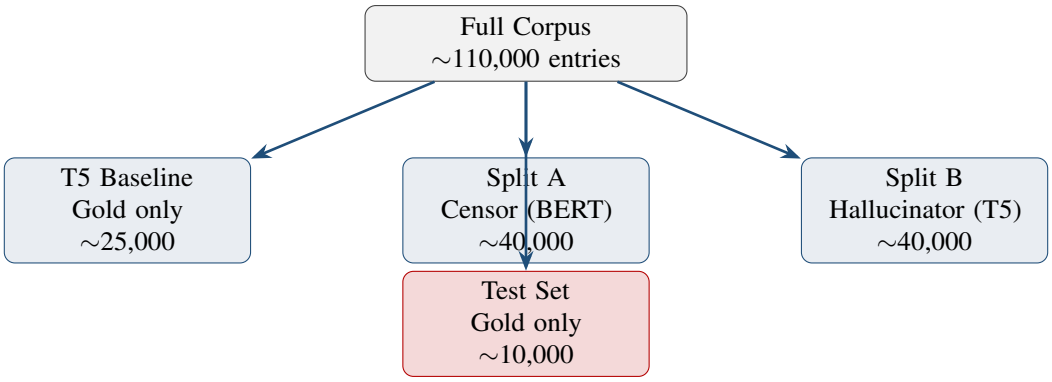

Figure 4: Final corpus partitioning. Split A $\cap$ Split B $= \emptyset$ (enforced by entry-ID disjointness).

12 Annotation Guidelines Summary

Seven Annotation Rules

1. **Tag the minimal span.** "John Smith" not "My name is John Smith."
2. **Prefer specific types.** `SSN` over `ID_NUMBER`; `EMAIL` over `OTHER_PII`.
3. **Format preservation.** Replacement must structurally match original (phone $\rightarrow$ phone format; date $\rightarrow$ same date format).
4. **Gender agreement.** Replacing a gendered name requires updating adjacent pronouns and honorifics.
5. **Cultural plausibility.** Replacement should plausibly belong to the same cultural/linguistic context as the original.
6. **No cross-contamination.** A replacement must not equal any original entity in the dataset (enforced by global registry).
7. **When in doubt, FLAG.** Flagged entries are escalated to the expert annotator and logged for IAA analysis.

13 References

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